

standard_competency_gap_analysis_method

Competency Gap Analysis Method

Purpose

This standard describes the accepted method for identifying competency gaps for a given target role, prior to curriculum or assessment design. A competency gap analysis is the first step in the overall curriculum design workflow, and its quality directly determines the relevance of everything produced downstream — a poorly conducted gap analysis will lead to modules and assessments that address the wrong competencies, or address competencies where no real gap actually existed.

Step 1: Identify the Target Role and Its Competency List

A competency gap analysis begins by identifying the target role precisely (for example, “Junior Backend Developer” rather than the broader and less precise “Developer”) and retrieving its associated competency list from the Backend Competency Framework or an equivalent approved framework. If the target role does not map cleanly to an existing framework, this should be flagged explicitly rather than proceeding with an improvised or approximate competency list, since doing so would undermine the traceability that the rest of this method depends on.

Step 2: Gather Evidence of Current Proficiency

Each competency identified in Step 1 is then compared against the current, demonstrated proficiency level of the individual or cohort being assessed. Acceptable forms of evidence include: results from prior formal assessments explicitly mapped to the same competency framework, patterns observed in code review history (such as recurring reviewer feedback on the same type of issue), and, where no stronger evidence exists, self-reported experience — though self-reported experience should be weighted as the weakest form of evidence available and should not be the sole basis for a significant curriculum decision when stronger evidence could reasonably be gathered instead.

Step 3: Record Gaps Only Where Evidence Supports Them

A gap is recorded only when there is reasonable evidence that a competency’s current demonstrated level falls below the target level expected for the role. Gaps should never be inferred purely from the absence of evidence for a given competency — absence of evidence should instead trigger a recommendation for direct, targeted assessment of that specific competency, rather than an assumed gap being recorded on the basis of missing information alone.

This distinction is treated as important enough to state explicitly because the two failure modes it prevents have different costs: assuming a gap exists without evidence tends to produce redundant training for competencies the individual may already possess, wasting time and potentially causing disengagement; assuming proficiency

without evidence risks missing a genuine skill deficiency that then surfaces later, at a point where the cost of addressing it (for example, after a defect reaches production) is significantly higher than it would have been during structured training.

Step 4: Document Each Recorded Gap Completely

Each recorded gap should reference the specific competency framework identifier (for example, “2.1 Relational Database Design”), the target proficiency level expected for the role, the currently observed proficiency level, and the specific evidence supporting that observed level. Gap analyses that omit supporting evidence for a recorded gap should not be used as the sole basis for subsequent curriculum design decisions, since a downstream curriculum designer or reviewer has no way to independently verify or challenge an unsupported gap claim.

Step 5: Hand Off to Curriculum Design

Once gaps are identified and fully documented per Step 4, they are handed off to curriculum design, which is expected to produce module outlines addressing each recorded gap individually, following the Module Outline Design Standard, with explicit gap-to-module traceability maintained throughout.

Handling Conflicting Evidence

If evidence gathered in Step 2 conflicts — for example, a prior formal assessment suggests proficiency at the target level, but recent code review history suggests otherwise — this conflict should be noted explicitly in the gap analysis rather than silently resolved in favor of one source over the other without explanation. In most cases, more recent evidence should be weighted more heavily than older evidence, since competency can both improve and regress over time, but the specific reasoning for how a conflict was resolved should always be recorded alongside the gap analysis itself.

Standard Owner

Curriculum Design Standards Committee.